Inverse Kinematics Wrist Math

EGR 445/545

GVSU

Plan for Today

- schedule talk
- quick review of inverse kinematics
- inverse kinematics wrist math
- LA 09
- project time

Announcements

- ~~HW 2 due 6/24~~
- ~~Quiz 3 also 6/24~~
  - ~~inverse kinematics~~
  - ~~one page of notes~~
  - ~~Python allowed~~
- ~~LA 10 dues 6/26~~
- ~~Exam 2 on 7/3~~
  - ~~inverse kinematics~~

See Spreadsheet

RB

Quick Review of Inverse Kinematics Concepts

- inverse kinematics is hard and we often seek ways to break problems into smaller pieces
- a fully algebraic solution is tough
- we can often break an industrial inv. kin. problem in half at the wrist
- knowing how to solve simpler 3DOF problems can help with these half problems
- our approach will combined splitting problems at the wrist with a geometric approach for the first half
- we have to know the orientation of the final link to use this approach
- we need to convert the customers problem statement to ${}^{0}_{tip}\mathbf{T}$ to use our approach

Where is the wrist?

- two options that are often the same:
  - the base of the last link
  - where the last 2-3 $\hat{Z}$ axes intersect

white 5DoF
FANUC

Many 6DoF robots
(yellow FANUC)

Key Concepts for Inverse Kinematics

- a proper problem statement for an industrial robot should include ${}^{0}_{tip}\mathbf{T}$
  - what would you need to formulate ${}^{0}_{tip}\mathbf{T}$?
- in general, # DOF = # of joints = # parameters needs for a valid problem statement

Inverse Kinematics “Algorithm”

- find ${}^{0}P_{wrist}$
  - if we have enough information
- draw a top view
  - what variables can we solve for?
- draw a side view
  - what frame do we need to draw the side view in?
  - apply geometric approach to the side view
- verify wrist location based on side view solution
- solve for the wrist angles
- verify final answer
  - reproduce ${}^{0}_{tip}\mathbf{T}$ based on forward kinematics

Big Idea:

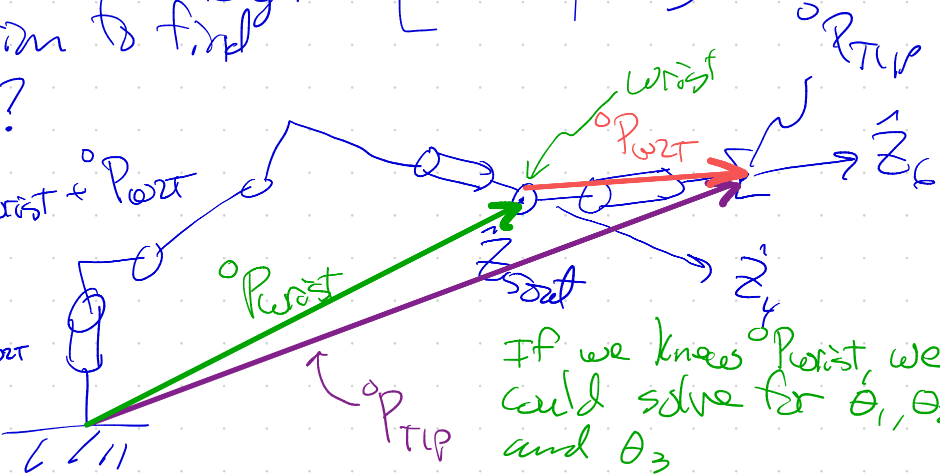
→ inverse kinematics of 5-6 DOF industrial robots would be nearly impossible if we don't split the problem in half at the wrist

Must be given ${}^{0}_{TIP}T = \left[\begin{array}{c|c} {}^{0}_{TIP}R & {}^{0}P_{TIP} \\ \hline 0\ 0\ 0 & 1 \end{array}\right]$

Do we have enough information to find ${}^{0}P_{wrist}$?

${}^{0}P_{TIP} = {}^{0}P_{wrist} + {}^{0}P_{w2T}$

${}^{0}P_{wrist} = {}^{0}P_{TIP} - {}^{0}P_{w2T}$



If we knew ${}^{0}P_{wrist}$, we could solve for θ_1, θ_2, and θ_3

QT Question

How do we find ${}^{0}P_{WRT}$?

→ d_6 is along $\hat{Z}_6$

$${}^{6}P_{WRT} = \begin{bmatrix} 0 \\ 0 \\ d_6 \end{bmatrix} \overset{?}{\Longrightarrow} {}^{0}P_{WRT} \, ?$$

from problem statement (${}^{0}_{TIP}T$)

from DH

$$\underbrace{{}^{0}P_{WRT} = {}^{0}_{6}R}_{\text{from problem statement}} \cdot \underbrace{{}^{6}P_{WRT}}_{\text{from DH}}$$

$${}^{0}_{6}R = {}^{0}_{TIP}R = \underbrace{{}^{0}_{TIP}T[0:3,\ 0:3]}_{\text{extract (slice) } {}^{0}_{6}R \text{ from } {}^{0}_{TIP}T}$$

extract (slice) ${}^{0}_{6}R$ from

${}^{0}_{TIP}T$

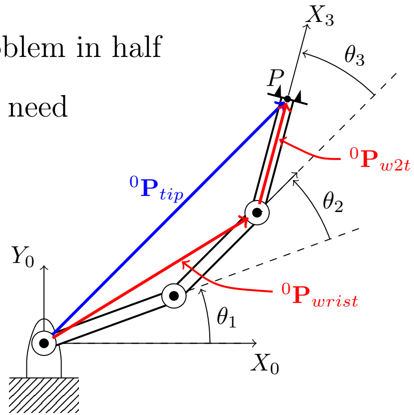
Finding the Wrist

LA 9

- essentially splits the problem in half
- what information do we need to find the wrist?

$$^0\mathbf{P}_{tip} = {}^0\mathbf{P}_{wrist} + {}^0\mathbf{P}_{w2t}$$

$$^0\mathbf{P}_{wrist} = {}^0\mathbf{P}_{tip} - {}^0\mathbf{P}_{w2t}$$



${}^{0}\mathbf{P}_{w2t}$

- how do we find the wrist-to-tip vector ${}^{0}\mathbf{P}_{w2t}$ in frame 0?
- ${}^{0}\mathbf{P}_{w2t} = {}^{0}\mathbf{P}_{tip} - {}^{0}\mathbf{P}_{wrist}$
- this is the relative displacement from O_N to the tip
    - the length should be the same in any frame
- typically:

$$
{}^{N}\mathbf{P}_{w2t} = \begin{bmatrix} a_{N-1} \\ 0 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 0 \\ 0 \\ d_N \end{bmatrix}
$$

${}^{0}\mathbf{P}_{w2t}$ continued

Having ${}^{N}\mathbf{P}_{w2t}$, how do we find ${}^{0}\mathbf{P}_{w2t}$?

- this is a relative displacement, so we just need to rotate it
    - we need ${}^{0}_{N}\mathbf{R}$, not ${}^{0}_{N}\mathbf{T}$
- how do we find ${}^{0}_{N}\mathbf{R}$?

From the problem statement

$$ {}^{0}_{tip}\mathbf{T} = \begin{bmatrix} & {}^{0}_{N}\mathbf{R} & & {}^{0}\mathbf{P}_{tip} \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

- ${}^{0}_{N}\mathbf{R} = {}^{0}_{tip}\mathbf{T}$ `[0:3,0:3]`
- ${}^{0}\mathbf{P}_{w2t} = {}^{0}_{N}\mathbf{R}\,{}^{N}\mathbf{P}_{w2t}$

Wrist Matrices

Problem: we are trying to find the wrist, but we don't know θ_4 yet

- the wrist is typically at O_4
  - yellow Fanuc is an example
- how can I use HT matrices to find O_4 without knowing θ_4?

$$ {}^{3}_{4}\mathbf{T} = {}^{3}_{W}\mathbf{T}\, {}^{W}_{4}\mathbf{T} $$

$$ {}^{0}_{tip}\mathbf{T} = {}^{0}_{1}\mathbf{T} \cdot {}^{1}_{2}\mathbf{T} \cdot {}^{2}_{3}\mathbf{T} \cdot {}^{3}_{W}\mathbf{T} \cdot {}^{W}_{4}\mathbf{T} \cdot {}^{4}_{5}\mathbf{T} $$

Wrist Matrices Continued

- ${}^{3}_{W}\mathbf{T} = {}^{3}_{4}\mathbf{T}\Big|_{\theta_4=0} = \mathtt{DH}(\alpha_3, a_3, 0, d_4)$
- ${}^{W}_{4}\mathbf{T} = {}^{3}_{4}\mathbf{T}\Big|_{\alpha_3=a_3=d_4=0} = \mathtt{DH}(0, 0, \theta_4, 0)$

Solving for Wrist Angles

Big Idea:

$$ {}^{W}_{tip}\mathbf{T}_{\text{numeric}} = {}^{W}_{tip}\mathbf{T}_{\text{symbolic}} $$

- Find ${}^{W}_{tip}\mathbf{T}_{\text{symbolic}}$ based on symbolic forward kinematics
- ${}^{W}_{tip}\mathbf{T}_{\text{numeric}}$ should be all hard numbers
- we have 12 equations (4x4 matrices)
    - pick options that lead to easy equations for θ_4, θ_5, and θ_6

Numeric Approach for ${}^{W}_{tip}\mathbf{T}$

- Find wrist location in frame {0}
- Find θ_1 from the top view
- Find θ_2 and θ_3 from the wrist position in frame {1} using the geometric approach
- Once we have θ_1, θ_2, and θ_3:

$$ {}^{0}_{W}\mathbf{T}_{\text{numeric}} = {}^{0}_{1}\mathbf{T} \cdot {}^{1}_{2}\mathbf{T} \cdot {}^{2}_{3}\mathbf{T} \cdot {}^{3}_{W}\mathbf{T} $$

$$ {}^{0}_{tip}\mathbf{T} = {}^{0}_{W}\mathbf{T} \cdot {}^{W}_{tip}\mathbf{T} $$

$$ {}^{W}_{tip}\mathbf{T}_{\text{numeric}} = {}^{0}_{W}\mathbf{T}^{-1} \cdot {}^{0}_{tip}\mathbf{T} $$

Quick Review: Finding the wrist in frame 0

$$^{0}P_{wrist} = {}^{0}P_{tip} - {}^{0}P_{w2t}$$

- what code do we need to find $^{0}P_{wrist}$?

Quick Review: Side View in Frame 1

- what does this mean?
- how do I find ${}^1P_{tip}$ or ${}^1P_{wrist}$?

Fig. 3.15 applied to the wrist

$${}^{3}_{4}\mathbf{T} = {}^{3}_{W}\mathbf{T} \cdot {}^{W}_{4}\mathbf{T}$$

Solving for wrist angles

$$ {}^{W}_{tip}\mathbf{T}_{numeric} = {}^{W}_{tip}\mathbf{T}_{symbolic} $$

- how do we find ${}^{W}_{tip}\mathbf{T}_{numeric}$?
- how do we find ${}^{W}_{tip}\mathbf{T}_{symbolic}$?

Inverse Kin. for a "Typical" Industrial Robot

- PUMA, Fanuc, simplified PUMA, …
- key characteristics:
  - last 2-3 $\hat{Z}$ axes intersect at the wrist
  - wrist moves in a plane set by θ_1
- approach:
  - find ${}^{0}P_{wrist}$
  - find θ_1 from top view of wrist location
  - find θ_2 and θ_3 from geometric approach on side view
  - probably in frame 1
  - verify wrist location: ${}^{0}_{W}\mathbf{T}$
  - find wrist angles from ${}^{W}_{tip}\mathbf{T}_{numeric} = {}^{W}_{tip}\mathbf{T}_{symbolic}$
  - verify full solution using forward kinematics

PUMA Wrist Angle Example

- `PUMA_inverse_kinematics_wrist_example.ipynb`
  - under `examples` on the learning activities `github` page

`https://github.com/ryanGT/learning_activities_EGR_445_545/blob/main/examples/PUMA_inverse_kinematics_wrist_example.ipynb`